31. The Discriminant Theorem for Orders of an Algebraic Number Field or Function Field

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Dedekind's discriminant theorem says, as is well known: a prime number enters the field discriminant if and only if it is divisible at least by the square of a prime ideal.

I show in what follows that this theorem remains valid for all finite orders of a finite algebraic number field, that is, for all such rings of algebraic integers which contain the ring of all rational integers, in the following form: a prime number p enters the discriminant of an order if and only if, in the decomposition valid in the order of the principal ideal (p) into greatest primary components, at least one proper primary ideal occurs, that is, one which is not a prime ideal.[1]

Since for the principal order, the system of all integers of the field, there are no further primary ideals except powers of prime ideals, Dedekind's theorem follows by specialization.

The proof of the theorem is reduced, by passing to the residue-class ring of the order modulo (p), to the decomposition theory of those rings which are of finite rank with respect to a field contained in the ring, that is, which form a commutative system of hypercomplex quantities with principal identity and coefficients from that same field. Here, in particular, the coefficient field is the residue-class field of the rational integers modulo p, and the decomposition of (p) corresponds in the residue-class ring to the decomposition of the zero ideal.

Thus if such a ring of finite rank is called completely reducible when its zero ideal can be represented as the least common multiple of finitely many prime ideals, the question is to find criteria for complete reducibility. As such a criterion one first obtains that, for a perfect field as coefficient domain, the ring is completely reducible if and only if the extension ring with algebraically closed coefficient domain is completely reducible. For this extension ring, however, the nonvanishing of the discriminant is very easily recognized as a necessary and sufficient criterion for complete reducibility;[2] and since the discriminant of the order modulo each prime number passes into the discriminant of the residue-class ring, the discriminant theorem is thereby proved.[3]

If one considers function fields instead of number fields, the order must now contain the ground domain of all rational integral functions, respectively, for algebraic functions of several indeterminates or for integral algebraic functions, the ground domain of all integral functionals with rational integral coefficients. In that case, the case of an imperfect coefficient domain can also occur in the residue-class ring, for example when, in the case of an integral functional domain, one takes the residue-class ring modulo a prime number. For an imperfect field as coefficient domain one must distinguish between prime ideals of the first and of the second kind according as the residue field modulo the prime ideal is an extension field of the first or of the second kind,[4] and correspondingly between complete reducibility of the first and of the second kind. The criteria stated above refer throughout to complete reducibility of the first kind, which alone occurs in the perfect case. Hence, in the most general case, the discriminant theorem says that a prime element of the ground domain enters the discriminant of an order if and only if, in the ideal decomposition of (p), at least one properly primary component or one prime ideal of the second kind occurs.

For the principal order this fact was first noticed in examples by Kronecker, after the Festschrift had still given, erroneously, the formulation valid for the principal order of a number field, though without a complete proof. For the principal order Ostrowski[5] gave a proof of the above discriminant theorem and further ramification theorems attached to it; at the same time a detailed survey of the literature is found there. Finally, the discriminant theorem for all orders of relative fields follows as a direct consequence, by passing to the quotient ring with respect to each prime ideal of the ground domain. The idea of this passage, which is known in the special case, was emphasized in principle by W. Krull;[6] a systematic theory of the quotient ring in the framework of more general investigations is given by H. Grell.[7]

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[1] Cf. E. Noether, *Abstrakter Aufbau der Idealtheorie in algebraischen Zahl- und Funktionenkörpern*, Math. Ann. 96 (1926), pp. 26–61. For all definitions reference is made to this paper, cited below as "Ideal Theory". In what follows, "order" always means "finite order", and "ring" a commutative ring.

[2] In a special case Hilbert argues similarly: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1896.

[3] Related investigations, also in the noncommutative case, but restricted to the principal order (maximal domain) and assuming rational integral number coefficients, are found in the recently published paper of A. Speiser, *Allgemeine Zahlentheorie*, Naturf. Gesellschaft Zürich 71 (1926), pp. 8–48. Cf. also the American literature cited there.

[4] In Steinitz's familiar formulation, J. f. M. 137. Cf. the note to §2, no. 4, of this paper.

[5] A. Ostrowski, Zur arithmetischen Theorie der algebraischen Größen, Nachr. Gött. Ges. d. Wissensch. 1919.

[6] Cf. his Danzig lecture, Jhrbr. d. D. Math. Ver. 34, p. 121.

[7] *Beziehungen zwischen den Idealen verschiedener Ringe* (to appear in Math. Annalen).

The uniform treatment of all orders given here, and of perfect and imperfect coefficient domains, rests, besides the use of ideal theory, on taking from the outset the theory of rings of finite rank, and hence the general concept of a finite extension, as the basis, and not the concept of an extension generated by a primitive element. The module basis of an order, together with the relations among its elements, therefore takes the place of the powers of a primitive element of the field and of the defining equation,[8] respectively of the powers of the fundamental form and of the fundamental equation.

Such a module basis passes, modulo every prime element of the ground domain, into a basis of the residue-class ring, whereas this need not hold even for the principal order when the basis consists of powers of a primitive element; nor for the basis consisting of powers of the fundamental form as soon as integral algebraic functions of several indeterminates are involved.[9] Expressed differently: the residue-class ring of the polynomial domain in one indeterminate with coefficients in the ground domain, modulo an arbitrary defining equation, or even modulo the fundamental equation, need not be isomorphic modulo p to the residue-class ring of the principal order modulo (p). The main difficulties in the customary presentations lie in this disappearance of the isomorphism. Since this auxiliary polynomial domain does not occur here, these difficulties are thereby avoided automatically. At the same time the additive decomposition theory of the residue-class ring can be understood as the equivalent of the multiplicative decomposition of the equation; here too, decomposition into relatively prime factors corresponds to an additive decomposition in the polynomial residue-class ring.

It should be noted that the discriminant theorem can represent only the first theorem of a general ramification theory of orders, just as in the principal order the discriminant theorem is accompanied by the more refined theory of the ramification ideal, which there permits a more precise determination of exponents. This determination of exponents, which rests on the power representation of the primary factors of the equations, can be interpreted as the determination of the length of a composition series from p to $q = p^t$, and will presumably occur in this form in the general ramification theory.

§1. Extension rings of fields

1. We first place in front a few remarks on arbitrary rings. If T is a ring and S a system of elements of T, then by the "ring derived from S in T" is meant the intersection of all rings in T which contain S. Correspondingly one defines the "ideal derived from S in T", or the "P-module derived from S in T", when P is a subring of T.[10]

If R is a subring of T and S a system of elements of T, the ring derived from R and S in T is denoted as the ring $R[S]$ arising by ring-adjunction of S to R. This ring consists of all polynomials in the elements of S with coefficients from R, where the equality and operation relations are fixed by the equality and operation relations in T. Because of these already fixed equality and operation relations it may suffice, in a special case, to form only a subsystem of all polynomials, for instance only all linear forms in S with coefficients from R. Such a special case occurs, for example (cf. no. 2), when R is a ring with identity element and S is a field with the same identity element: all linear combinations $\sum c_i\gamma_i$, where c_i belongs to S and γ_i to R, form the ring $R[S]$.

But when only the ring R is given, and in addition a system S of elements not contained in R with equality relations, or also mutual operation relations, one can also construct the extension ring $R[S]$ arising by adjunction of S to R. One shows that there is always a ring containing R and S, namely the ring formally consisting of all polynomials in S with coefficients from R, with the usual operation rules fixed. For the definition of equality, aside from the operation rules, full freedom is still possible, except for the restriction that the equality definitions valid in R and S must be included. Thus it is necessary only that those polynomials be set equal which, by virtue of equality in R and S and by virtue of the operations, can be formed from equal elements of R and S in the same way. The ring T so constructed, containing R and S, is then identical with the ring derived in T from R and S.[11]

2. Extension rings of fields. From now on assume rings $\mathfrak{R}$ with identity element, containing a field P as a subring, and therefore over-rings of fields. The identity element e of $\mathfrak{R}$ is then also the identity element of P, and the ring is a P-module.

If $\overline{P}$ is an overfield of P, the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ arising by ring-adjunction of $\overline{P}$ to $\mathfrak{R}$ is to be

[8] This point of view is closely connected with Dedekind's view of higher complex numbers: Zur Theorie der aus n Haupteinheiten gebildeten komplexen Größen, Gött. Nachr. 1885.

[9] Cf. Ostrowski, loc. cit., "irregularity of the first kind".

[10] Cf. Ideal Theory, §1.

[11] Cf. Steinitz's construction of the quotient field.

explained as follows.

It is assumed that the set-theoretic intersection $[\mathfrak{R}, \overline{P}]$ of $\mathfrak{R}$ and $\overline{P}$ is P, and further that between the elements of $\mathfrak{R}$ and $\overline{P}$ not belonging to P no operation relations exist. This assumption can always be achieved by replacing $\overline{P}$ by an equivalent extension field of P, or also by replacing $\mathfrak{R}$ by an equivalent extension ring.[12]

As elements of $\overline{\mathfrak{R}}$ one declares all formally formed linear combinations

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s},$$

where the $\bar{c}_i$ run through all elements of $\overline{P}$ and the γ_i through all elements of $\mathfrak{R}$.[13] The definition is to be unique; that is, if the γ are replaced by elements equal to them in $\mathfrak{R}$, and the $\bar{c}$ by elements equal to them in $\overline{P}$, then the element in $\overline{\mathfrak{R}}$ is also to be declared equal.

As sum one defines, uniquely in the sense that replacing a summand by an equal one in the equality definition to be fixed gives the same result,

$$\sum \bar{c}_i\gamma_i + \sum \bar{d}_i\gamma_i = \sum(\bar{c}_i + \bar{d}_i)\gamma_i.$$

Here zeros may occur among the $\bar{c}$ and $\bar{d}$, whereby formally the same γ_i occur.

As product one defines, in the same sense uniquely,

$$\sum \bar{c}_i\gamma_i \cdot \sum \bar{d}_k\gamma_k = \sum \bar{c}_i\bar{d}_k\gamma_i\gamma_k.$$

For the equality definition it is declared that, if $\gamma_{i_1}, \ldots, \gamma_{i_s}$ are linearly independent with respect to P, then two elements

$$\bar{c}_{i_1}\gamma_{i_1} + \cdots + \bar{c}_{i_s}\gamma_{i_s} \quad \text{and} \quad \bar{d}_{i_1}\gamma_{i_1} + \cdots + \bar{d}_{i_s}\gamma_{i_s}$$

are equal if and only if $\bar{c}_{i_k}$ is equal to $\bar{d}_{i_k}$ in $\overline{P}$. Expressed differently: elements of $\mathfrak{R}$ linearly independent with respect to P are declared to be linearly independent with respect to $\overline{P}$.

By the definitions of sum and product the equality definition for all elements of $\overline{\mathfrak{R}}$ is thereby given. Indeed their expressibility by linearly independent elements from $\mathfrak{R}$ follows. For from a relation

$$\mu = c_1\gamma_1 + \cdots + c_s\gamma_s \quad \text{in } \mathfrak{R}$$

one obtains, by multiplication with $\bar{b} = be$ according to the product definition,

$$\bar{b}\mu = \bar{b}e \cdot (c_1\gamma_1 + cdots + c_s\gamma_s) = \bar{b}c_1\gamma_1 + \cdots + \bar{b}c_s\gamma_s,$$

and hence, by the sum definition, the expressibility of all elements by linearly independent elements from $\mathfrak{R}$.

Consequently, if all elements have been expressed by linearly independent elements from $\mathfrak{R}$, the equality definition is equality of coefficients in $\overline{P}$. It therefore agrees, for elements belonging simultaneously to $\mathfrak{R}$ and $\overline{\mathfrak{R}}$ or to $\overline{P}$ and $\overline{\mathfrak{R}}$, with the equality valid there; whereas for the remaining elements of $\overline{\mathfrak{R}}$ the equality valid in $\mathfrak{R}$ and $\overline{P}$ says nothing beyond the uniqueness required in the definition, because of the intersection assumption and the further condition.[14] Similarly, the requirement that sum and product be unique entails no further relations, since equal elements may be assumed coefficientwise equal, and hence formally give the same sum and product expressions.

One now verifies at once that all axioms of the ring operation are fulfilled; thus the extension ring $\mathfrak{R}[\overline{P}]$ generated by ring-adjunction always exists.

3. Ideals in $\mathfrak{R}$ and $\overline{\mathfrak{R}}$. If $\mathfrak{a}$ is an ideal in $\mathfrak{R}$, then the module product $\overline{\mathfrak{R}}\mathfrak{a}$ is an ideal $\bar{\mathfrak{a}}$ in $\overline{\mathfrak{R}}$, the extension ideal of $\mathfrak{a}$. At the same time $\bar{\mathfrak{a}}$ is the ideal derived from $\mathfrak{a}$ in $\overline{\mathfrak{R}}$; it consists of all linear combinations $\sum \bar{c}_i a_i$, the sum in each case being over finitely many elements, where the a_i run through all elements of $\mathfrak{a}$ and the $\bar{c}_i$ through all elements of $\overline{P}$.

[12] Two extension fields of P are called "equivalent", following Steinitz, if they can be placed in isomorphism with one another in such a way that P corresponds elementwise to itself. Equivalent extension rings of P are to be defined correspondingly. By introducing new symbols one can always produce equivalent extensions; this freedom in the use of equivalent extensions is essential for what follows.

[13] In general, elements from $\mathfrak{R}$ or $\overline{\mathfrak{R}}$ are denoted by Greek letters, and elements from P or $\overline{P}$ by Latin letters.

[14] Without this assumption there would be at least one element $\bar{c}$ belonging to $\overline{P}$ but not to P, for which $\bar{c} = \bar{c}e = \gamma$ would hold. Then, by the equality definition in $\overline{\mathfrak{R}}$, e and γ would indeed be linearly independent with respect to P, but dependent with respect to $\overline{P}$. Because of the relations between elements of $\overline{P}$ and $\mathfrak{R}$, the above equality definition would then not be possible; think, for instance, of the lowering of degree of a finite extension field when the ground field is replaced by an intermediate field. The above extension is, by contrast, characterized by the possibility of the appearance of new zero divisors. Thus, for example, the residue-class ring of the polynomial domain with rational coefficients P modulo $x^2 + 1$ is a ring without zero divisors, isomorphic to the field $P(\sqrt{-1})$, whereas on passing to the polynomial domain with coefficients from $P(\sqrt{-1})$ zero divisors occur, corresponding to $(x + i)(x - i) = x^2 + 1$, although no factor is divisible by $x^2 + 1$. The corresponding statement holds, for example, for every finite number field; this is the conception first expressed by Dedekind for the higher complex numbers. Cf. on this point §6, no. 4, at the end.

If $\overline{\mathfrak{b}}$ is an ideal in $\overline{\mathfrak{R}}$, then the intersection $[\overline{\mathfrak{b}}, \mathfrak{R}]$ is an ideal $\mathfrak{b}$ in $\mathfrak{R}$, the contraction ideal of $\overline{\mathfrak{b}}$.[15] Every ideal $\mathfrak{a}$ of $\mathfrak{R}$ is a contraction ideal, namely of its extension ideal $\overline{\mathfrak{a}}$. Indeed, every element a of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ is of the form $\sum \bar{c}_i a_i$, where the a_i may be assumed linearly independent with respect to P or $\overline{P}$. Thus the elements a, a_i become linearly dependent in $\overline{\mathfrak{R}}$, and hence, by the equality definition, also in $\mathfrak{R}$; therefore a belongs to $\mathfrak{a}$. Since the converse also holds, $\mathfrak{a} = [\mathfrak{R}, \overline{\mathfrak{a}}]$.

If $\overline{\mathfrak{p}}$ is a prime ideal in $\overline{\mathfrak{R}}$, then $\mathfrak{p} = [\mathfrak{R}, \overline{\mathfrak{p}}]$ is a prime ideal in $\mathfrak{R}$. For, by the second isomorphism theorem,[16] the residue-class ring $\mathfrak{R}/\mathfrak{p}$ is isomorphic to a subring of $\overline{\mathfrak{R}}/\overline{\mathfrak{p}}$, and hence is a ring without zero divisors. Namely, there is the ring isomorphism

$$(\mathfrak{R}, \overline{\mathfrak{p}})/\overline{\mathfrak{p}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{p}}],$$

where $(\mathfrak{R}, \overline{\mathfrak{p}})$ denotes the sum, the greatest common divisor, of the modules $\mathfrak{R}$ and $\overline{\mathfrak{p}}$.

§2. Rings of finite rank. Representation as a direct sum

1. Rings of finite rank. If $\mathfrak{R}$ is a finite P-module, then $\mathfrak{R}$ is of finite rank, say k, with respect to P: that is, there are k, and no more, elements in $\mathfrak{R}$ linearly independent with respect to P, and every system of k elements of $\mathfrak{R}$ linearly independent with respect to P consequently forms a module basis of $\mathfrak{R}$. If a P-module $\mathfrak{B}$ of finite rank is divisible by another $\mathfrak{A}$ of the same finite rank, $\mathfrak{B} = O(\mathfrak{A})$, then they are therefore identical.

From now on assume that $\mathfrak{R}$ has finite rank n with respect to P. Then every P-module in $\mathfrak{R}$ has finite rank. Thus in every proper chain of divisors or multiples of P-modules in $\mathfrak{R}$, the rank of each module must be larger, respectively smaller, than the rank of the immediately preceding one; the chains break off after finitely many steps. In particular, the "double-chain theorem" holds for the system of all ideals in $\mathfrak{R}$.

It follows from this[17] that a prime ideal $\mathfrak{p}$ of $\mathfrak{R}$ has no proper divisor different from the identity ideal; hence the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo a prime ideal different from the identity ideal is a field. If $\mathfrak{q}$ is a primary ideal belonging to $\mathfrak{p}$, that is, $\mathfrak{q} = O(\mathfrak{p})$ and $\mathfrak{p}^\rho = O(\mathfrak{q})$, then the residue-class ring $\mathfrak{R}/\mathfrak{q}$ is a primary ring, that is, a ring in which a power, here in any case already the ρ-th, of every zero divisor vanishes; and $\mathfrak{R}/\mathfrak{q}$ contains only zero divisors and units. The latter follows immediately from the fact that every ideal not divisible by $\mathfrak{p}$ is relatively prime to $\mathfrak{p}$, and hence also to $\mathfrak{q}$.

From the double-chain theorem and from the existence of the identity element it follows further[18] that every ideal $\mathfrak{a}$ of $\mathfrak{R}$ can be represented uniquely as a product of finitely many pairwise relatively prime primary ideals. Because of relative primeness this product is equal to the least common multiple and corresponds to a representation of the residue-class ring $\mathfrak{R}/\mathfrak{a}$ as a direct sum of primary rings.[19] That is, $\mathfrak{R}/\mathfrak{a}$ is the greatest common divisor of these rings, and the sum representation of every element of $\mathfrak{R}/\mathfrak{a}$ is unique.

2. Representation of rings of finite rank as direct sums of primary rings. Suppose that the representation of the zero ideal of $\mathfrak{R}$ is given by

$$(0) = \mathfrak{q}_1\mathfrak{q}_2 \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \mathfrak{q}_2, \ldots, \mathfrak{q}_r],$$

and let the corresponding representation as a direct sum be

$$\mathfrak{R} = \mathfrak{R}_1 + \mathfrak{R}_2 + \cdots + \mathfrak{R}_r, \qquad \mathfrak{R}_i\mathfrak{R}_k = (0)\ (i \neq k), \qquad \mathfrak{R}_i^2 = \mathfrak{R}_i, \qquad \mathfrak{R}\mathfrak{R}_i = \mathfrak{R}_i.$$

Then $\mathfrak{R}_i$ is an ideal of $\mathfrak{R}$, namely

$$\mathfrak{R}_i = \mathfrak{q}_1 \cdots \mathfrak{q}_{i-1}\mathfrak{q}_{i+1} \cdots \mathfrak{q}_r = [\mathfrak{q}_1, \ldots, \mathfrak{q}_{i-1}, \mathfrak{q}_{i+1}, \ldots, \mathfrak{q}_r],$$

$$\mathfrak{q}_i = \mathfrak{R}_1 + \cdots + \mathfrak{R}_{i-1} + \mathfrak{R}_{i+1} + \cdots + \mathfrak{R}_r,$$

and $\mathfrak{R}_i$ is isomorphic to the residue-class ring $\mathfrak{R}/\mathfrak{q}_i$: to each element γ_i of $\mathfrak{R}_i$ corresponds the class represented by γ_i in $\mathfrak{R}/\mathfrak{q}_i$.[20] If $\mathfrak{a}$ is any ideal of $\mathfrak{R}$, then the distributive law gives

$$\mathfrak{a} = \mathfrak{R}\mathfrak{a} = \mathfrak{R}_1\mathfrak{a} + \cdots + \mathfrak{R}_r\mathfrak{a} = \mathfrak{a}_1 + \cdots + \mathfrak{a}_r;$$

[15] In H. Grell's notation, in the work cited in note 6 on p. 83 of the original.

[16] Ideal Theory, §4, no. 3. The theorem is there stated only for ideals of the same ring; the same consideration shows its validity in the above form for arbitrary ideals $\overline{\mathfrak{a}}$ of $\overline{\mathfrak{R}}$. The ring homomorphism $\overline{\mathfrak{R}} \to \overline{\mathfrak{R}}/\overline{\mathfrak{a}}$ induces, for the subring $\mathfrak{R}$ of $\overline{\mathfrak{R}}$, the ring homomorphism $\mathfrak{R} \sim (\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}}$. Under this homomorphism all and only the elements of $[\mathfrak{R}, \overline{\mathfrak{a}}]$ correspond to the zero element, and hence the ring isomorphism $(\mathfrak{R}, \overline{\mathfrak{a}})/\overline{\mathfrak{a}} \simeq \mathfrak{R}/[\mathfrak{R}, \overline{\mathfrak{a}}]$ follows. The classes can be represented on the right and on the left by the same elements of $\mathfrak{R}$, for at each step of the proof the elements are assigned to the classes represented by them.

[17] Ideal Theory, §7, no. 2.

[18] Ideal Theory, §7, no. 3, Theorem V.

[19] Ideal Theory, §4, no. 5. What is added in the present compilation is found scattered in many places in the literature.

[20] Ideal Theory, §4, no. 5.

here, since $\mathfrak{a}_i = \mathfrak{R}_i\mathfrak{a} = \mathfrak{R}\mathfrak{a}_i$, the components $\mathfrak{a}_i$ are ideals both in $\mathfrak{R}_i$ and in $\mathfrak{R}$. As ideals, the $\mathfrak{a}_i$, like the $\mathfrak{R}_i$ themselves, are finite P-modules; because the sum is direct, their ranks add to the rank of $\mathfrak{a}$, respectively of $\mathfrak{R}$.

If the sum representation of the identity element is given by

$$e = \varepsilon_1 + \cdots + \varepsilon_r, \qquad \varepsilon_i\varepsilon_k = 0\ (i \neq k), \qquad \varepsilon_i^2 = \varepsilon_i, \qquad \varepsilon_i \equiv e \pmod{\mathfrak{q}_i},$$

then the representation of every element γ of $\mathfrak{R}$ is obtained uniquely as

$$\gamma = \gamma e = \gamma\varepsilon_1 + \cdots + \gamma\varepsilon_r = \gamma_1 + \cdots + \gamma_r,$$

where the component $\gamma_i = \gamma\varepsilon_i$ is an element of $\mathfrak{R}_i$; at the same time $\mathfrak{R}_i = \mathfrak{R}\varepsilon_i$. From the "orthogonality relations" one obtains

$$\gamma\delta = (\gamma_1\varepsilon_1 + \cdots + \gamma_r\varepsilon_r)(\delta_1\varepsilon_1 + \cdots + \delta_r\varepsilon_r) = \gamma_1\delta_1\varepsilon_1 + \cdots + \gamma_r\delta_r\varepsilon_r.$$

Thus $\gamma_i\delta_i$ is the i-th component of the product, and similarly $\gamma_i + \delta_i$ is the i-th component of the sum, facts which also follow from the homomorphism from $\mathfrak{R}$ to $\mathfrak{R}_i$.

The ring $\mathfrak{R}_i$ contains a subfield $P_i = P\varepsilon_i$ isomorphic to P and is of finite rank with respect to P_i; this rank is equal to the rank possessed by $\mathfrak{R}_i$ as a P-module. For for every element c of P, one has $c\varepsilon_i \equiv ce \equiv c \pmod{\mathfrak{q}_i}$, and hence $c\varepsilon_i \neq 0$ as soon as $c \neq 0$; the homomorphism between P and P_i becomes an isomorphism. Further, every linear dependence with respect to P of elements of $\mathfrak{R}_i$ passes, by multiplication with ε_i, into such a dependence with respect to P_i; conversely, because $P_i = P\varepsilon_i$, every relation with respect to P_i is at the same time a relation with respect to P. Thus the ranks with respect to P and P_i agree.

3. Direct sum representation of extension rings. If

$$\mathfrak{R} = \mathfrak{R}_1 + \cdots + \mathfrak{R}_r,$$

then for $\overline{\mathfrak{R}}$ a representation

$$\overline{\mathfrak{R}} = \overline{\mathfrak{R}}_1 + \cdots + \overline{\mathfrak{R}}_r$$

holds, where $\overline{\mathfrak{R}}_i$ denotes the extension ideal of $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$, at the same time the ring derived from $\mathfrak{R}_i$ in $\overline{\mathfrak{R}}$. Indeed, from $(0) = \mathfrak{q}_1 \cdots \mathfrak{q}_r$, multiplication by $\overline{\mathfrak{R}}$ gives the product representation $(0) = \overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_r$, where the $\overline{\mathfrak{q}}_i$ are again pairwise relatively prime. Hence one obtains a direct-sum representation for $\overline{\mathfrak{R}}$, and since the i-th component of this representation is given by

$$\overline{\mathfrak{q}}_1 \cdots \overline{\mathfrak{q}}_{i-1}\, \overline{\mathfrak{q}}_{i+1} \cdots \overline{\mathfrak{q}}_r,$$

it is equal to the extension ideal of $\mathfrak{R}_i$. But by §1, no. 3, this extension ideal is identical with the extension ring $\mathfrak{R}_i[\overline{P}_i]$ formed from $\mathfrak{R}_i$ with respect to $\overline{P}_i = \overline{P}\varepsilon_i$; therefore in extensions it suffices to extend the individual components. In general $\overline{\mathfrak{R}}_i$ will then no longer be a primary ring.

4. Prime ideals of the first and second kind. By no. 1, the residue-class ring $\mathfrak{R}/\mathfrak{p}$ modulo every prime ideal different from the identity ideal is a field. This field possesses a subfield $(P) = (P, \mathfrak{p})/\mathfrak{p} = P/[P, \mathfrak{p}] \simeq P$ isomorphic to P, since $\mathfrak{p}$ can contain no nonzero element from P; hence (P) consists of all and only the classes which can be represented by elements of P.[21] Thus $\mathfrak{R}/\mathfrak{p}$ is of finite rank with respect to (P), hence is a finite algebraic extension field. According as this extension is of the first or second kind,[22] $\mathfrak{p}$ is to be called a prime ideal of the first or of the second kind.

§3. Completely reducible rings

1. The ring $\mathfrak{R}$ is called *completely reducible* if, in the product representation of its zero ideal (§2, no. 2), all primary components become prime ideals; or – what is identical with this by §2, nos. 1 and 2 – if $\mathfrak{R}$ can be represented as a direct sum of finitely many fields. The completely reducible ring is called completely reducible of the first kind if all prime ideals are of the first kind (§2, no. 4); in the opposite case it is completely reducible of the second kind.

[21] Cf. the note to §1, no. 3, on the second isomorphism theorem.

[22] In the familiar terminology of Steinitz: an algebraic extension is called of the first kind if every element is a zero of a prime function which, in a suitable extension field, decomposes into distinct linear factors; otherwise it is of the second kind.

If P is a perfect field, then only complete reducibility of the first kind occurs; in particular this is always the case for an algebraically closed field. In what follows $\overline{P}$ denotes specifically the algebraic closure derived from P, and $\overline{\mathfrak{R}}$ is equal to $\mathfrak{R}[\overline{P}]$.

2. **Theorem.** *The ring $\mathfrak{R}$ is completely reducible of the first kind if and only if the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ with algebraically closed $\overline{P}$ is completely reducible.*

The proof proceeds in three steps.

2a. From complete reducibility of $\overline{\mathfrak{R}}$ – more generally, of any extension ring – there follows complete reducibility (of the first or second kind) of $\mathfrak{R}$. For by hypothesis the zero ideal of $\overline{\mathfrak{R}}$ has a representation as the least common multiple of prime ideals,

$$(0) = [\overline{\mathfrak{p}}_1, \dots, \overline{\mathfrak{p}}_s].$$

By contraction one obtains in $\mathfrak{R}$ the representation of the zero ideal

$$(0) = [[\mathfrak{R}, \overline{\mathfrak{p}}_1], \dots, [\mathfrak{R}, \overline{\mathfrak{p}}_s]],$$

and the ideals $[\mathfrak{R}, \overline{\mathfrak{p}}_i]$ are prime ideals by §1, no. 3.

2b. From complete reducibility of the first kind of $\mathfrak{R}$ there follows complete reducibility of the first kind of $\overline{\mathfrak{R}}$, and more generally complete reducibility of the first kind of every intermediate ring between $\mathfrak{R}$ and $\overline{\mathfrak{R}}$.

Since, by §2, no. 3, a representation of $\mathfrak{R}$ as a direct sum gives a corresponding representation of $\overline{\mathfrak{R}}$ such that each component $\overline{\mathfrak{R}}_i$ becomes the extension ring $\mathfrak{R}_i[\overline{P}_i]$ – where $\overline{P}_i$ is an extension field isomorphic to $\overline{P}$ of the subfield P_i of the component $\mathfrak{R}_i$ isomorphic to P – it suffices to consider the individual component $\mathfrak{R}_i$. Thus, without loss of generality, $\mathfrak{R}$ may be assumed to be a field, indeed a finite extension field of the first kind over its subfield P. Then $\mathfrak{R}$ is generated by adjoining a primitive element of the first kind to P; equivalently, $\mathfrak{R}$ is isomorphic to the residue-class ring $P[x]/f(x)$, where $f(x)$ is a prime function of the first kind in the polynomial domain $P[x]$ in one indeterminate x. If Q is an extension field of P, then $Q[x]/f(x)$ is isomorphic to $\mathfrak{R}[Q]$; for, in passing from $P[x]$ to $Q[x]$, the rank of the residue-class ring is preserved, equal to the degree of $f(x)$, so that precisely the construction of the extension ring given in §1, no. 2 is involved. In $Q[x]$ the polynomial $f(x)$ decomposes into pairwise coprime prime functions of the first kind, each of which generates a prime ideal in $Q[x]$; hence $Q[x]/f(x)$, and therefore by the isomorphism also $\mathfrak{R}[Q]$, is completely reducible of the first kind. Since this holds for every extension field Q, in particular also for the algebraically closed field $\overline{P}$, 2b is proved.

2c. If $\mathfrak{R}$ is completely reducible of the second kind, then $\overline{\mathfrak{R}}$ is not completely reducible.

As in 2b, it suffices to suppose that $\mathfrak{R}$ is a field, and indeed a finite extension field of the second kind over P. It must be shown that in the representation of $\overline{\mathfrak{R}}$ as a direct sum of primary rings at least one proper primary component occurs. For this it is enough to exhibit a non-zero element of $\overline{\mathfrak{R}}$ some power of which vanishes. For if $\bar{\beta} \neq 0$, at least one component $\bar{\beta}_i$ must be non-zero; and if $\bar{\beta}^{\rho} = 0$, then $\bar{\beta}_i^{\rho} = 0$ (§2, no. 2, the homomorphism from $\overline{\mathfrak{R}}$ to $\overline{\mathfrak{R}}_i$), and hence $\overline{\mathfrak{R}}_i$ is not a field.

Let γ be an element of the second kind in $\mathfrak{R}$, of exponent $f \geq 1$, and let

$$F(\gamma) = \gamma^{kp^f} + c_1 \gamma^{(k-1)p^f} + \dots + c_k = 0$$

be the equation of lowest degree satisfied by γ over P, where p denotes the characteristic of P. The elements

$$e, \gamma, \gamma^2, \dots, \gamma^{kp^f - 1}$$

of $\mathfrak{R}$ are therefore linearly independent over P and consequently remain, by the definition of equality, linearly independent over $\overline{P}$ in $\overline{\mathfrak{R}}$. If one now sets

$$G(\gamma) = \gamma^{kp^{f-1}} + \sqrt[p]{c_1}\, \gamma^{(k-1)p^{f-1}} + \dots + \sqrt[p]{c_k},$$

then $G(\gamma)$ is a non-zero element of $\overline{\mathfrak{R}}$; for, since $p > 1$ and $f \geq 1$, the exponent kp^{f-1} is always smaller than kp^f, and no dependence among the powers of γ occurring in $G(\gamma)$ can exist. But $G(\gamma)^p = F(\gamma)$ and hence vanishes. Thus 2c is proved.

From 2a and 2c it follows that complete reducibility of the first kind of $\overline{\mathfrak{R}}$ implies complete reducibility of the first kind of $\mathfrak{R}$; 2b gives the converse. This proves the theorem.

3. **Corollary.** *If $\mathfrak{R}$ is completely reducible of the first kind, then $\overline{\mathfrak{R}}$ is a direct sum of rings of rank one, hence of fields isomorphic to $\overline{P}$. Such a decomposition into rings of rank one already exists in a ring $\mathfrak{R}[Q]$, where Q is a finite extension field of P.*

For by 2b the ring $\overline{\mathfrak{R}}$ becomes a direct sum of fields which, as finite extensions of subfields isomorphic to $\overline{P}$, coincide with those subfields because $\overline{P}$ is algebraically closed. Thus one obtains

$$\overline{\mathfrak{R}} = \overline{P}\bar{e}_1 + \cdots + \overline{P}\bar{e}_n, \qquad e = \bar{e}_1 + \cdots + \bar{e}_n;$$

the components $\bar{e}_i$ of e at the same time form a linearly independent module basis of $\overline{\mathfrak{R}}$.

If the $\bar{e}_i$ are represented, according to §1, no. 2, as linear combinations of elements γ of $\mathfrak{R}$ with coefficients in $\overline{P}$, then the system of all these coefficients determines a finite extension field Q of P. Hence the $\bar{e}_i$ already belong to $\mathfrak{R}[Q]$, and from the module-basis property one obtains

$$\mathfrak{R}[Q] = Q\bar{e}_1 + \cdots + Q\bar{e}_n.$$

Thus already in $\mathfrak{R}[Q]$ there is a decomposition into rings of rank one, and in every intermediate ring between $\mathfrak{R}[Q]$ and $\overline{\mathfrak{R}}$ the indecomposable components arise as extension rings of the $Q\bar{e}_i$.[23]

§4. Matrix representation, traces, and discriminants for rings of finite rank

In this and the next paragraph the assumptions on the underlying ring are slightly more general than before, so as also to include orders in number fields and function fields. What is involved is a general and uniform formulation of essentially known facts.

Assumption. $\mathfrak{R}$ is an extension ring of a domain $\mathfrak{Z}$; the identity element of $\mathfrak{R}$ already lies in $\mathfrak{Z}$. For every $\mathfrak{Z}$-module in $\mathfrak{R}$ – in particular for $\mathfrak{R}$ itself – there exists at least one finite module basis consisting of elements linearly independent over $\mathfrak{Z}$. Besides $\alpha_1, \dots, \alpha_s$, the elements $\beta_1, \dots, \beta_s$ form a linearly independent module basis of the same $\mathfrak{Z}$-module if and only if the transformation determinant is a unit of $\mathfrak{Z}$.

The assumption is plainly fulfilled for the rings of finite rank considered up to now, where $\mathfrak{Z}$ is the field P. It is also fulfilled for the orders in number fields and function fields, where $\mathfrak{Z}$ is the ring of rational integers, respectively the functional domain.[24]

1. *Matrix rings homomorphic to* $\mathfrak{R}$. Let $\alpha_1, \dots, \alpha_s$ be a linearly independent $\mathfrak{Z}$-module basis of the ideal $\mathfrak{a}$, and let γ be any element of $\mathfrak{R}$. Then $\gamma\alpha_i$ also belongs to $\mathfrak{a}$, and there are equations with coefficients in $\mathfrak{Z}$:

$$\gamma\alpha_i = c_{i1}\alpha_1 + \cdots + c_{is}\alpha_s, \qquad i = 1, \dots, s.$$

In matrix form, if $(\tau_1, \dots, \tau_s)$ denotes the one-row matrix formed from the elements in question,

$$(\gamma\alpha_1, \dots, \gamma\alpha_s) = (\alpha_1, \dots, \alpha_s)\begin{pmatrix} c_{11} & \cdots & c_{1s} \\ \vdots & & \vdots \\ c_{s1} & \cdots & c_{ss} \end{pmatrix} = (\alpha_1, \dots, \alpha_s)C.$$

As γ runs through all elements of $\mathfrak{R}$, the totality of matrices C assigned by means of the basis α_i forms a ring $R_{\mathfrak{a}}$ homomorphic to $\mathfrak{R}$; $R_{\mathfrak{a}}$ is isomorphic to the residue-class ring $\mathfrak{R}/((0) : \mathfrak{a})$, where $(0) : \mathfrak{a}$ denotes the ideal quotient of the zero ideal by $\mathfrak{a}$. For to the difference $\beta - \gamma$ there is assigned the difference $B - C$, and the same holds for products, since

$$(\beta\gamma\alpha_1, \dots, \beta\gamma\alpha_s) = (\gamma\alpha_1, \dots, \gamma\alpha_s)B = (\alpha_1, \dots, \alpha_s)CB.$$

To the elements c of $\mathfrak{Z}$ there correspond in particular the diagonal matrices cE. Exactly those elements γ for which $\gamma\mathfrak{a}$ is the zero ideal are assigned the zero matrix.

A different basis $\bar{\alpha}_1, \dots, \bar{\alpha}_s$ of the same ideal generates a matrix ring isomorphic to $R_{\mathfrak{a}}$, indeed equivalent to it:

$$R_{\bar{\mathfrak{a}}} = P^{-1}R_{\mathfrak{a}}P.$$

For if $(\bar{\alpha}_1, \dots, \bar{\alpha}_s) = (\alpha_1, \dots, \alpha_s)P$, then the corresponding matrices C' and C satisfy $C' = P^{-1}CP$.

2. *Ideal classes and representation classes.* Two ideals $\mathfrak{a}$ and $\mathfrak{b}$ of $\mathfrak{R}$ belong to the same ideal class if they are isomorphic as $\mathfrak{R}$-modules, i.e. if their elements can be put into a one-to-one correspondence such

[23]The smallest such extension field Q may be characterized as the compositum of the Galois fields determined by the components $\mathfrak{R}_i = \mathfrak{R}e_i$ of $\mathfrak{R}$. Namely, choose $f_i(x)$ as in 2b so that $P[x]/f_i(x)$ is isomorphic to the component $\mathfrak{R}_i$, and let $Q^{(i)}$ be the Galois extension field lying in $\overline{P}$ which just suffices to split $f_i(x)$ into linear factors. Then Q is the compositum of all $Q^{(i)}$. The splitting of $f_i(x)$ into linear factors corresponds to a representation of $Q^{(i)}[x]/f_i(x)$ as a direct sum of fields of rank one; $Q^{(i)}$ is the smallest extension field in which this happens. By isomorphism the same holds for $\mathfrak{R}_i[Q^{(i)}\bar{e}_i]$ and hence for $\mathfrak{R}[Q]$.

[24]Cf. *Ideal Theory*, §3. Elements of $\mathfrak{Z}$ are denoted by Latin letters.

that difference and multiplication by the same element of $\mathfrak{R}$ correspond. Two rings $R_{\mathfrak{a}}$ and $R_{\mathfrak{b}}$ produced by matrix representation as in no. 1 belong to the same representation class if they are equivalent, i.e. if there is a unimodular matrix P such that

$$R_{\mathfrak{b}} = P^{-1} R_{\mathfrak{a}} P.$$

Ideal classes and representation classes correspond one-to-one. Different module bases of the same ideal give equivalent matrix rings; and different ideals in the same ideal class likewise give equivalent matrix rings, because corresponding basis elements yield the same linear coefficients when multiplied by any $\gamma \in \mathfrak{R}$. Conversely, if $R_{\mathfrak{a}}$ and $R_{\mathfrak{b}}$ belong to the same representation class, the transformation matrix supplies a basis of $\mathfrak{b}$ in which to the element $a = c_1\alpha_1 + \cdots + c_s\alpha_s$ of $\mathfrak{a}$ there corresponds $b = c_1\beta_1 + \cdots + c_s\beta_s$ of $\mathfrak{b}$; equality of the matrix rings then gives $\gamma a \leftrightarrow \gamma b$. The class of the unit ideal is called the principal class.

3. *Trace of an element with respect to a class.* Let C be the matrix assigned to the element γ of $\mathfrak{R}$ in $R_{\mathfrak{a}}$. Passing from $R_{\mathfrak{a}}$ to an equivalent ring shows that $|C|$, and more generally $|tE - C|$, where t is an indeterminate, is an invariant of the representation class. Hence it is also an invariant of the ideal class. The coefficient of $(-t^{s-1})$ in $|tE - C|$ is to be called the trace of γ with respect to the class:

$$S_{(\mathfrak{a})}(\gamma) = c_{11} + c_{22} + \cdots + c_{ss}.$$

The constant coefficient $\pm|C|$ is called the norm of γ with respect to the class.

For a fixed class the traces $S_{(\mathfrak{a})}(\gamma)$ form a $\mathfrak{Z}$-module homomorphic to $\mathfrak{R}$ viewed as a $\mathfrak{Z}$-module. Indeed, by the ring homomorphism proved in no. 1,

$$S_{(\mathfrak{a})}(\beta - \gamma) = S_{(\mathfrak{a})}(\beta) - S_{(\mathfrak{a})}(\gamma), \qquad S_{(\mathfrak{a})}(c\beta) = cS_{(\mathfrak{a})}(\beta).$$

If $\mathfrak{R}$ is a ring without zero divisors, then the trace of an element is defined independently of the class; one may simply speak of the trace. The same holds for the norm. For every non-zero ideal then has the rank of the ring, and every basis of an ideal is simultaneously a basis of the unit ideal in the quotient field.

4. *Discriminant of an ideal with respect to a class.* It is not the discriminant itself, but only the principal ideal derived from it in $\mathfrak{Z}$, that becomes an invariant of ideal and class. Let $\omega_1, \ldots, \omega_s$ and $\bar{\omega}_1, \ldots, \bar{\omega}_s$ be two linearly independent $\mathfrak{Z}$-module bases of an ideal $\mathfrak{t}$ of $\mathfrak{R}$. The determinants

$$|S_{(\mathfrak{a})}(\omega_i\omega_j)|, \qquad |S_{(\mathfrak{a})}(\bar{\omega}_i\bar{\omega}_j)|$$

differ only by the square of a unit of $\mathfrak{Z}$. The same is true for the two determinants of the basis products; and because of the module homomorphism, the same linear relations hold among the traces as among the products, so that under the cogredient transformation the determinant acquires the same quadratic factor.

Every determinant $|S_{(\mathfrak{a})}(\omega_i\omega_j)|$ is called a discriminant of $\mathfrak{t}$ with respect to the class $(\mathfrak{a})$, determined up to the square of a unit; the principal ideal derived from it in $\mathfrak{Z}$ is called the discriminant ideal. If $\mathfrak{R}$ is a ring without zero divisors, then the discriminant of an ideal is independent of the class used.

§5. Direct sums of classes

1. *Direct sum of the ideal class or representation class.* If the ideal $\mathfrak{a}$ is a direct sum, $\mathfrak{a} = \mathfrak{b} + \mathfrak{c}$, then from the isomorphism it follows that every ideal in the class is also a direct sum, $\bar{\mathfrak{a}} = \bar{\mathfrak{b}} + \bar{\mathfrak{c}}$. Here $\bar{\mathfrak{b}}$ is respectively isomorphic to $\mathfrak{b}$ and $\bar{\mathfrak{c}}$ to $\mathfrak{c}$, the ideals being regarded as $\mathfrak{R}$-modules; one may therefore speak of the direct sum of the ideal class and of its component classes.

If a ring $R_{\mathfrak{a}}$ in a representation class is a direct sum of subrings which are at the same time ideals in $R_{\mathfrak{a}}$, then by isomorphism the same is true for every ring $R_{\mathfrak{b}}$ in the representation class, and the components are equivalent rings. Thus one may speak of the direct sum of the representation class and of the component classes. The direct sum of the ideal class corresponds to the direct sum of the representation class; in this sense one speaks of a direct sum of the class.[25]

Let $\beta_1, \ldots, \beta_m$ be a basis of $\mathfrak{b}$ and $\gamma_1, \ldots, \gamma_n$ a basis of $\mathfrak{c}$; because the sum is direct, the β's and γ's together form a linearly independent basis of $\mathfrak{a}$. If $R_{\mathfrak{a}}$ is the matrix ring generated by this basis, then the matrix in $R_{\mathfrak{a}}$ assigned to any element δ of $\mathfrak{R}$ has the form

$$\begin{pmatrix} D^{\mathfrak{b}} & 0 \\ 0 & D^{\mathfrak{c}} \end{pmatrix}.$$

[25] The question to what extent all direct sums of representation classes can be produced by direct sums of ideal classes is not taken up here.

The systems of matrices containing only the first, respectively only the second, block form ideals $R^{\mathfrak{b}}$ and $R^{\mathfrak{c}}$ in $R_{\mathfrak{a}}$; their product vanishes, and $R_{\mathfrak{a}} = R^{\mathfrak{b}} + R^{\mathfrak{c}}$ is a direct sum. The assignment of $D^{\mathfrak{b}}$ to the corresponding block matrix in $R^{\mathfrak{b}}$ gives an isomorphism with the ring $R_{\mathfrak{b}}$ generated from $\mathfrak{b}$, and similarly for $\mathfrak{c}$.

2. *Behavior of trace and discriminant under direct sums of classes.* If $\mathfrak{a} = \mathfrak{b} + \mathfrak{c}$ and one uses the block matrix to determine the trace $S_{(\mathfrak{a})}(\delta)$, one reads off: the trace of an element taken with respect to a class is the sum of the traces taken with respect to the component classes. Because the product $\mathfrak{bc}$ vanishes, one further reads off: if β is an element of $\mathfrak{b}$, then $S_{(\mathfrak{a})}(\beta) = S_{(\mathfrak{b})}(\beta)$; the trace of β with respect to the class $(\mathfrak{a})$ equals its trace with respect to the class $(\mathfrak{b})$.

It follows that the discriminant ideal of the component ideal $\mathfrak{b}$, taken with respect to the class $(\mathfrak{a})$, equals the discriminant ideal of $\mathfrak{b}$ taken with respect to the component class $(\mathfrak{b})$. Finally, the discriminant ideal of $\mathfrak{a}$, taken with respect to $(\mathfrak{a})$, equals the product of the discriminant ideals of the components, taken with respect to the component classes. For with the basis adapted to the direct sum the discriminant matrix is block diagonal:

$$\begin{vmatrix} S_{(\mathfrak{b})}(\beta_i\beta_j) & 0 \\ 0 & S_{(\mathfrak{c})}(\gamma_i\gamma_j) \end{vmatrix} = |S_{(\mathfrak{b})}(\beta_i\beta_j)|\,|S_{(\mathfrak{c})}(\gamma_i\gamma_j)|.$$

3. *Passage to extension rings.* Let $\mathfrak{X}$ be an extension ring without zero divisors of $\mathfrak{Z}$ such that the intersection of $\mathfrak{R}$ and $\mathfrak{X}$ is $\mathfrak{Z}$; let the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\mathfrak{X}]$ obtained by adjoining $\mathfrak{X}$ be defined as a ring of the same rank. If $\mathfrak{a}$ and $\mathfrak{t}$ are ideals in $\mathfrak{R}$ and $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{t}}$ their extension ideals in $\overline{\mathfrak{R}}$, then the discriminant ideal of $\overline{\mathfrak{t}}$, taken with respect to $(\overline{\mathfrak{a}})$, is also the extension ideal in $\mathfrak{X}$ of the discriminant ideal of $\mathfrak{t}$ taken with respect to $(\mathfrak{a})$. For every basis of $\mathfrak{a}$ or $\mathfrak{t}$ is also an $\mathfrak{X}$-module basis of $\overline{\mathfrak{a}}$ or $\overline{\mathfrak{t}}$. In particular, the discriminant ideal of $\overline{\mathfrak{R}}$ with respect to the principal class is the extension ideal of that of $\mathfrak{R}$ with respect to the principal class.

§6. Discriminant criterion for complete reducibility of the first kind

From now on $\mathfrak{R}$ is again assumed to be an extension ring of a field P, and $\overline{P}$ is an extension field of P. The non-vanishing of the discriminant will turn out to be the necessary and sufficient condition for complete reducibility of the first kind. By §5, no. 3, the discriminant ideal of $\mathfrak{R}$ can be determined by means of the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$; and by §5, no. 2, it suffices to restrict attention to the primary rings of which $\overline{\mathfrak{R}}$ is additively composed.

1. *Special P-module bases.* If $\mathfrak{a}$ and $\mathfrak{t}$ are ideals of $\mathfrak{R}$ and $\mathfrak{a}$ is divisible by $\mathfrak{t}$, then every basis of $\mathfrak{a}$ can be completed, by adjoining elements, to a linearly independent module basis of $\mathfrak{t}$. If $\mathfrak{R}$ is a primary ring, $\mathfrak{p}$ the prime ideal belonging to the zero ideal, and ρ the exponent, so that $\mathfrak{p}^\rho = (0)$ and $\mathfrak{p}^{\rho-1} \neq (0)$, then a special P-module basis of $\mathfrak{R}$ is obtained by successively completing a basis of $\mathfrak{p}^{\rho-1}$ to one of $\mathfrak{p}^{\rho-2}$, and in general a basis of $\mathfrak{p}^i$ to one of $\mathfrak{p}^{i-1}$, with $\mathfrak{R}$ counted as $\mathfrak{p}^0$.

With such a basis one can show: if $\mathfrak{R}$ is primary, then the trace, taken with respect to the principal class, of every element divisible by the corresponding prime ideal $\mathfrak{p}$ vanishes. Namely, let

$$\eta_1, \ldots, \eta_s$$

be such a special basis, ordered by the powers of $\mathfrak{p}$. From $\pi \equiv 0 \pmod{\mathfrak{p}}$ it follows that multiplication by π moves each basis block into the next higher power of $\mathfrak{p}$; hence the assigned matrix has only zeros in the diagonal, and the trace of π vanishes.[26]

2. *Discriminant ideals of primary rings with algebraically closed subring P.* If the zero ideal of $\mathfrak{R}$ is a prime ideal and P is algebraically closed, then $\mathfrak{R}$ is of rank one and hence identical with P (§3, no. 3); the identity element e may be taken as module basis. Thus

$$S_{(\mathfrak{R})}(e^2) = S_{(\mathfrak{R})}(e) = e,$$

and the discriminant ideal is the unit ideal.

The discriminant ideal of a proper primary ring with algebraically closed subring P is the zero ideal. If one uses the special module basis of no. 1 – or merely completes any basis of $\mathfrak{p}$ to a basis of $\mathfrak{R}$ – the first basis element may be taken to be the identity element e, and all basis products different from e^2 are divisible by $\mathfrak{p}$; their trace vanishes. Thus the discriminant vanishes as a determinant of size at least two in which all entries except $S(e^2)$ vanish.

[26] The matrix representation furnished by such a basis has the block-triangular form belonging to the ideal factors. Since in complete reducibility the indecomposable components have no ideal different from the zero and unit ideals, such a non-trivial representation cannot occur in the representation class of these component ideals.

3. **Theorem.** *The ring $\mathfrak{R}$ of finite rank over a subfield P is completely reducible of the first kind if and only if its discriminant ideal is the unit ideal in P; in other words, if and only if the discriminant, determined up to the square of a unit, is non-zero.*

Complete reducibility of the first kind is, by §3, no. 2, identical with the assertion that the extension ring $\overline{\mathfrak{R}} = \mathfrak{R}[\overline{P}]$ is completely reducible, that is, that its indecomposable components are fields isomorphic to $\overline{P}$. By §5, no. 3, the discriminant ideal of $\mathfrak{R}$ and at the same time that of $\overline{\mathfrak{R}}$ is either the zero ideal or the unit ideal; and by §5, no. 2, the discriminant ideal of $\overline{\mathfrak{R}}$ is the product of the discriminant ideals of its components, taken with respect to these components. The individual discriminant ideals arise from those considered in no. 2 by replacing, in each case, the field $P_i = \overline{P}e_i$ isomorphic to $\overline{P}$ by $\overline{P}$ itself. Hence, if $\overline{\mathfrak{R}}$ is completely reducible, its discriminant ideal, being a product of unit ideals, is the unit ideal; whereas if $\overline{\mathfrak{R}}$ is not completely reducible, at least one component has discriminant ideal equal to the zero ideal, and consequently the discriminant ideal of $\overline{\mathfrak{R}}$ is zero. This proves the theorem.

4. *Representation of trace, norm, and discriminant by conjugate elements in the case of complete reducibility of the first kind.* Let, by §3, no. 3, in the case of complete reducibility of the first kind,

$$\mathfrak{R}[Q] = Qe_1 + \cdots + Qe_n$$

be a representation as a direct sum of rings of rank one, and let Q be the smallest extension field in which such a representation is possible. The n components of $\mathfrak{R}$ are then given by K_ie_i, where each K_i is a subfield of Q and $\mathfrak{R}$ is homomorphic to K_i.

Taking $e_1, \ldots, e_n$ as module basis, every element

$$\gamma = c_1e_1 + \cdots + c_ne_n$$

is assigned the diagonal matrix with diagonal entries $c_1, \ldots, c_n$. Consequently

$$S(\gamma) = c_1 + \cdots + c_n, \qquad N(\gamma) = c_1 \cdots c_n,$$

where $S(\gamma)$ is the trace and $N(\gamma)$ the norm with respect to the principal class. If $\alpha_1, \ldots, \alpha_n$ is a module basis of $\mathfrak{R}$ consisting of elements of $\mathfrak{R}$, and

$$\alpha_j = \alpha_j^{(1)}e_1 + \cdots + \alpha_j^{(n)}e_n,$$

then

$$|S(\alpha_i\alpha_j)| = \left|\alpha_i^{(k)}\right|^2.$$

That is, the discriminant is represented as the square of the determinant of the conjugate basis elements. If in particular $\mathfrak{R}$ itself is a field, hence an extension field of the first kind of the subfield P, then the homomorphisms from $\mathfrak{R}$ to the K_i become isomorphisms, and the K_i are the conjugate fields with respect to P in the Galois extension field Q. The representation of trace, norm, and discriminant then becomes the familiar representation by conjugate elements. It should again be emphasized that $\mathfrak{R}$ is assumed only as a field equivalent to the K_i, not as a field already lying in the Galois extension field Q.

§7. The discriminant theorem for the orders of an algebraic number field or function field

1. The number fields and function fields under consideration all fit the following field type, with a fixed notion of integral elements. Let $\mathfrak{o}$ be a principal ideal domain without zero divisors and with identity (the ring of rational integers, the polynomial domain in one indeterminate with coefficients in a field, or the functional domain), and let Ω be its quotient field. Let K be a finite extension of the first kind of Ω, and let $\mathfrak{O}$ be the ring of all elements of K integral over $\mathfrak{o}$. Every subring of $\mathfrak{O}$ which contains $\mathfrak{o}$ is called an order; $\mathfrak{O}$ itself is called the principal order.

Every $\mathfrak{o}$-module in $\mathfrak{O}$, in particular every ideal of an order and the order itself, has a linearly independent module basis over $\mathfrak{o}$. If K has degree n over Ω, it is enough to restrict attention to orders of rank n over $\mathfrak{o}$. Every order $\mathfrak{T}$ is therefore a ring satisfying the assumptions of §4 and §5, and indeed a ring without zero divisors; hence the discriminant $D_{\mathfrak{T}}$ of the order is uniquely determined up to the square of a unit of $\mathfrak{o}$ as a factor. This discriminant is also, up to a unit of Ω, identical with the discriminant of the field K viewed as an Ω-module, and is therefore non-zero, since K was assumed to be an extension of the first kind (§6, no.

3). At the same time the discriminant has the representation by the square of the determinant of conjugate basis elements given in §6, no. 4.

The ideal theory in $\mathfrak{T}$ is given by the theorem: in $\mathfrak{T}$, every non-zero ideal can be uniquely represented as a product of finitely many pairwise coprime primary ideals whose corresponding prime ideals have no proper divisor different from the unit ideal.

2. *Passage from the order to rings of finite rank over a field.* Let p be a prime element of $\mathfrak{o}$, so that the principal ideal derived from p in $\mathfrak{o}$ is a prime ideal $\mathfrak{o}p$ of $\mathfrak{o}$ different from the zero and unit ideals; let $\mathfrak{T}p$ denote correspondingly the principal ideal derived from p in $\mathfrak{T}$. The residue-class ring

$$\mathfrak{R} = \mathfrak{T}/\mathfrak{T}p$$

becomes a ring of rank n over a subfield P isomorphic to the field $\mathfrak{o}/\mathfrak{o}p$; $\mathfrak{T}$ is homomorphic to $\mathfrak{T}/\mathfrak{T}p$.

That $\mathfrak{R}$ contains a subfield isomorphic to $\mathfrak{o}/\mathfrak{o}p$ follows from the second isomorphism theorem: the subring $(\mathfrak{o}, \mathfrak{T}p)/\mathfrak{T}p$ is isomorphic to $\mathfrak{o}/[\mathfrak{o}, \mathfrak{T}p] = \mathfrak{o}/\mathfrak{o}p$. Let $\alpha_1, \ldots, \alpha_n$ be a linearly independent module basis of $\mathfrak{T}$ over $\mathfrak{o}$, and let (α_i) be the residue classes determined by these elements modulo $\mathfrak{T}p$. Any linear dependence between the (α_i) over P would arise from a congruence

$$c_1\alpha_1 + \cdots + c_n\alpha_n \equiv 0 \pmod{\mathfrak{T}p},$$

hence from a relation $c_1\alpha_1 + \cdots + c_n\alpha_n = py$ with $y \in \mathfrak{T}$. From the representation of y in the basis α_i and the uniqueness of this representation, it follows that every c_i is divisible by p; consequently the element (c_i) of P vanishes. Thus the (α_i) form a basis of $\mathfrak{R}$ over P.

Under the homomorphism the discriminant $D_{\mathfrak{T}}$ of $\mathfrak{T}$ passes to the discriminant of $\mathfrak{R}$, and the ideal decomposition of $\mathfrak{T}p$ passes to the decomposition of the zero ideal of $\mathfrak{R}$. If

$$\mathfrak{T}p = \mathfrak{q}_1 \cdots \mathfrak{q}_s = [\mathfrak{q}_1, \ldots, \mathfrak{q}_s]$$

is the representation of $\mathfrak{T}p$ in $\mathfrak{T}$, then it becomes, under the homomorphism,

$$(0) = [(\mathfrak{q}_1), \ldots, (\mathfrak{q}_s)]$$

in $\mathfrak{R}$. By the first isomorphism theorem, $\mathfrak{R}/(\mathfrak{q}_i)$ is isomorphic to $\mathfrak{T}/\mathfrak{q}_i$; hence $\mathfrak{q}_i$ and $(\mathfrak{q}_i)$ are simultaneously proper primary ideals or prime ideals, and the $(\mathfrak{q}_i)$ are pairwise coprime by homomorphism.

3. **Discriminant theorem.** *A prime element p of $\mathfrak{o}$ divides the discriminant $D_{\mathfrak{T}}$ of an order $\mathfrak{T}$ if and only if, in the decomposition of the principal ideal $\mathfrak{T}p$ into pairwise coprime primary components in $\mathfrak{T}$, at least one proper primary component or at least one prime ideal of the second kind occurs.*

If the residue field $\mathfrak{o}/\mathfrak{o}p$ is a perfect field, then every prime element p which divides $D_{\mathfrak{T}}$ – and only such a p – has at least one proper primary component. If $\mathfrak{o}/\mathfrak{o}p$ is perfect and $\mathfrak{T}$ is chosen to be the principal order $\mathfrak{O}$, then p divides the discriminant if and only if it is divisible by at least the square of a prime ideal of $\mathfrak{O}$.

For, by the correspondence between the ideal decomposition of $\mathfrak{T}p$ in $\mathfrak{T}$ and that of the zero ideal in $\mathfrak{R} = \mathfrak{T}/\mathfrak{T}p$, the occurrence of a proper primary component or of a prime ideal of the second kind in $\mathfrak{T}p$ means that $\mathfrak{R}$ is not completely reducible of the first kind. By §6, no. 3, this is the case if and only if the discriminant of $\mathfrak{R}$ vanishes, which by the homomorphism in no. 2 says precisely that $D_{\mathfrak{T}}$ is divisible by p.[27]

§8. The discriminant theorem for the orders of relative fields

If instead of the principal ideal domain $\mathfrak{o}$ one already takes as the starting ring the principal order of a number field or function field – more generally, a multiplication ring, i.e. a ring in which the ideal theory of the principal order holds – then in place of the discriminant $D_{\mathfrak{T}}$ there appears the discriminant ideal of $\mathfrak{T}$ with respect to $\mathfrak{o}$; the discriminant theorem transfers completely by means of the following considerations.

1. Let $\mathfrak{o}$ be a multiplication ring, that is, a ring without zero divisors and with identity, in which every ideal different from the zero and unit ideals can be represented uniquely as a power product of prime ideals, and where these prime ideals have no proper divisor different from the unit ideal. Let $\Omega, K, \mathfrak{O}, \mathfrak{T}$ be defined as in §7, no. 1; then in $\mathfrak{T}$ the ideal theory stated there holds.

[27]Example of the occurrence of prime ideals of the second kind: let $\mathfrak{o}$ be the derived functional domain, Ω its quotient field, and let K be the extension field obtained by adjoining $\sqrt{x}$. Consider the order $\mathfrak{T}$ with module basis $1, \sqrt{x}$ over $\mathfrak{o}$. Its discriminant is $\begin{vmatrix} 1 & \sqrt{x} \\ 1 & -\sqrt{x} \end{vmatrix}^2$. The principal ideal $\mathfrak{T}2$ remains a prime ideal in $\mathfrak{T}$, but one of the second kind, since $\mathfrak{T}/\mathfrak{T}2$ is isomorphic to $P[t]/(t^2 - x)$, where P is the corresponding residue field. In contrast, $\mathfrak{T}\sqrt{x}$ is proper primary. If one replaces one indeterminate by two, one obtains examples that are generated over P only after adjoining two elements.

In particular, if a multiplication ring $\mathfrak{o}$ has only one prime ideal different from the zero and unit ideals, then $\mathfrak{o}$ is a principal ideal domain. For if one chooses an element p whose principal ideal is divisible by exactly the first power of this prime ideal, then p generates the prime ideal itself; the power-product representation then implies that all other ideals too are principal.

2. *Trace and discriminant ideal.* The trace of every element γ of $\mathfrak{T}$ is defined as the trace of γ in K, where K is regarded as an Ω-module. Every system of n elements of K linearly independent over Ω may be used as a basis for producing the trace. If $\alpha_1, \ldots, \alpha_n$ is such a system, then, since K is assumed to be an extension of the first kind,

$$|S(\alpha_i \alpha_j)|$$

is a non-zero element of Ω, indeed the square of the determinant of the conjugate basis elements. From the integral closedness of $\mathfrak{o}$ in Ω it follows that this determinant is an element of $\mathfrak{o}$ as soon as all α_i belong to $\mathfrak{O}$.

Let $\tau_1, \ldots, \tau_n$ run through all systems of n elements of $\mathfrak{T}$, and form, for every such system, the determinant $|S(\tau_i \tau_j)|$. The ideal in $\mathfrak{o}$ derived from the totality of these determinants is called the discriminant ideal of the order $\mathfrak{T}$ with respect to $\mathfrak{o}$. If in particular the order has a linearly independent module basis $\alpha_1, \ldots, \alpha_n$, then the determinant $|S(\alpha_i \alpha_j)|$ may be taken as a basis of the discriminant ideal; the definition agrees with that of §7, no. 1.

3. *Passage to the quotient ring.* If $\mathfrak{a}$ is any ideal of $\mathfrak{o}$, the quotient ring $\mathfrak{T}_{\mathfrak{a}}$ is defined as the totality of elements of the quotient field of $\mathfrak{T}$ whose denominator is an element of $\mathfrak{T}$ prime to $\mathfrak{a}$. Apart from the zero and unit ideals, the ring $\mathfrak{T}_{\mathfrak{a}}$ has no prime ideals other than the extension ideals of the prime ideals belonging to $\mathfrak{a}$. There is a one-to-one correspondence, with isomorphism of residue-class rings, between the non-trivial ideals of $\mathfrak{T}_{\mathfrak{a}}$ and those ideals of $\mathfrak{T}$ whose corresponding prime ideals are among those belonging to $\mathfrak{a}$; moreover the zero and unit ideals correspond one-to-one.

The same considerations apply to the multiplication ring $\mathfrak{o}$. If in particular $\mathfrak{p}$ is a prime ideal, then $\mathfrak{o}_{\mathfrak{p}}$ has only one prime ideal different from the zero and unit ideals; since the residue-class rings are isomorphic, $\mathfrak{o}_{\mathfrak{p}}$ is, with $\mathfrak{o}$, a multiplication ring, and by no. 1 it is therefore a principal ideal domain. The basis of the prime ideal derived from $\mathfrak{p}$ becomes a prime element p of $\mathfrak{o}_{\mathfrak{p}}$. The extension ideal of any ideal $\mathfrak{c}$ of $\mathfrak{o}$ is generated by the primary component of $\mathfrak{c}$ belonging to $\mathfrak{p}$, and is therefore equal to $(p)^{\alpha}$ or to the unit ideal according as $\mathfrak{p}$ occurs in $\mathfrak{c}$ – to the α-th power – or does not occur in $\mathfrak{c}$. If the extensions of two ideals in all quotient rings $\mathfrak{o}_{\mathfrak{p}}$ agree, then the two ideals are identical.

4. *The discriminant ideal under passage to the quotient ring.* Let $\mathfrak{p}$ be a prime ideal of $\mathfrak{o}$ and put $\mathfrak{P} = \mathfrak{T}\mathfrak{p}$ for the extension ideal in $\mathfrak{T}$. The quotient ring $\mathfrak{T}_{\mathfrak{P}}$ possesses a module basis consisting of linearly independent elements over the principal ideal domain $\mathfrak{o}_{\mathfrak{p}}$; at the same time $\mathfrak{T}_{\mathfrak{P}}$ is a finite $\mathfrak{p}$-order in the extension field K of the quotient field of $\mathfrak{o}_{\mathfrak{p}}$. Thus in $\mathfrak{T}_{\mathfrak{P}}$, with respect to $\mathfrak{o}_{\mathfrak{p}}$, the discriminant theorem (§7, no. 3) holds.

The discriminant $D_{\mathfrak{T}_{\mathfrak{P}}}$ is then a basis of the extension ideal, taken in $\mathfrak{o}_{\mathfrak{p}}$, of the discriminant ideal of $\mathfrak{T}$ with respect to $\mathfrak{o}$. Namely, choose elements $\alpha_1, \ldots, \alpha_n$ of $\mathfrak{T}$ as an $\mathfrak{o}_{\mathfrak{p}}$-module basis of $\mathfrak{T}_{\mathfrak{P}}$; then $|S(\alpha_i \alpha_j)|$ belongs to the discriminant ideal, and every determinant $|S(\tau_i \tau_j)|$ in the quotient ring is divisible by $|S(\alpha_i \alpha_j)|$.

5. **General discriminant theorem.** *A prime ideal* $\mathfrak{p}$ *of a multiplication ring* $\mathfrak{o}$ *occurs in the discriminant ideal of an order with respect to* $\mathfrak{o}$ *if and only if, in the ideal decomposition of* $\mathfrak{T}\mathfrak{p}$ *into pairwise coprime primary components in* $\mathfrak{T}$*, at least one proper primary component or at least one prime ideal of the second kind occurs.*

From 3 and 4 it follows that a prime ideal $\mathfrak{p}$ of $\mathfrak{o}$ occurs in the discriminant ideal of $\mathfrak{T}$ if and only if the prime element p of $\mathfrak{o}_{\mathfrak{p}}$ divides the discriminant $D_{\mathfrak{T}_{\mathfrak{P}}}$. But this is identical with saying that the residue-class ring $\mathfrak{T}_{\mathfrak{P}}/\mathfrak{T}_{\mathfrak{P}}p$ is not completely reducible of the first kind; and since this residue-class ring is isomorphic to $\mathfrak{T}/\mathfrak{T}\mathfrak{p}$, the general discriminant theorem is proved.[28]

As a specialization to relative discriminants of number fields one obtains: a prime ideal $\mathfrak{p}$ of the principal order of a number field occurs in the relative discriminant of an extension field if and only if $\mathfrak{p}$ is divisible in the principal order of the extension field by the square of a prime ideal.

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[28]When $\mathfrak{o}$ is the principal order of an algebraic number field, this gives agreement with the relative discriminant defined by Hilbert, which is known to agree with Hecke's definition.